

REVIEW ARTICLE

Current status and research progress in architectural programming: A comparative analysis between China and other countries



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Abstract Architectural programming is of significant importance in improving urban planning efficiency, enhancing the sustainability of the construction industry, and supporting the development of social infrastructure. However, due to its broad scope and complex content, research in this area exhibits diverse and scattered characteristics. To comprehensively understand the current status, progress, and future directions of architectural programming research, and to provide recommendations for both China and other countries, this study conducted a comparative review of relevant literature using a mixed method approach of bibliometrics, latent Dirichlet allocation (LDA) topic modeling, and systematic review. The study summarized the differences between China and other countries in six topics within the field of architectural programming: (1) intelligent and digital methods, (2) sustainability and practices, (3) project management and decision support, (4) urban planning and existing development, (5) social infrastructure and public services, and (6) architectural education and talent cultivation. This study provides an overview and comparative analysis on architectural programming, offering important reference for future research and practice, and promoting the overall development of the construction industry.

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1. Introduction

Architectural programming, also known as briefing (Kelly and Duerk, 2002), constitutes a pivotal phase in the early stages of architectural projects (Pena and Parshall, 2012). It primarily involves the process of gathering, analyzing, and integrating information to make decisions. It delineates the goals, conditions, and scope of a project, establishes the requirements of clients, users, experts, and other stakeholders (Hershberger, 2015), and continuously influences the subsequent aspects of the project, such as design, implementation, delivery, and management (Gray and Larson, 2011; McGeorge and Zou, 2012; O'Reilly, 1987). In recent years, with the development of socio-economic conditions (Chan et al., 2018), the acceleration of urbanization (Vaz and Nijkamp, 2015), and the continuous pursuit of quality of life (Schaffers et al., 2011), architectural projects face challenges such as slow advancement in informatization and automation (Maskuriy et al., 2019), increased security risks (Kardes et al., 2013), and irrational allocation of resources (Bowen et al., 2010). Concurrently, the rapid development of new-generation information technologies such as Artificial Intelligence (AI), Big Data, and the Internet of Things (IoT) is driving the transformation and upgrading of the entire architectural industry (Balasubramanian et al., 2021; Smallwood and Allen, 2023), thereby influencing the evolution of architectural programming (Zhuang et al., 2023).

Against this backdrop, countries worldwide have introduced a series of policies to promote the construction and management of the entire life cycle of the construction industry (Charef et al., 2022). For example, the United States places more emphasis on the safety, economic benefits, and sustainable development of construction projects, actively advocating green building and LEED certification systems (Azhar et al., 2011; Olawumi and Chan, 2018; Teizer et al., 2010). The United Kingdom prioritizes digital design, low-carbon, and sustainable architectural

strategies, promoting the sustainable development of the construction industry through policy guidance and incentive measures (Alwan et al., 2017; Dadhich et al., 2015; Iacovidou et al., 2021). China focuses more on intelligent construction and the integration of the entire industry chain, including design, construction, and operation, proposing a series of strategies and policy measures such as "Digital China" and "Smart Construction" to promote high-quality development in the construction industry (Ding et al., 2022; Qian et al., 2017; Wang et al., 2020). Under the impetus of these policies, architectural programming, as an essential link in addressing many challenges, has received high attention from governments and industries and has become a significant direction for policy adjustments in various countries (Abe et al., 2023; Lee et al., 2022b; Tang et al., 2015). It not only holds great potential in multidisciplinary fields (Oliveira and Marco, 2017; Xiang et al., 2021) but also plays a crucial role in the development of cities and societies (Gibson and Gebken, 2003; Watson et al., 2016).

Although the importance of architectural programming in practice is increasingly evident, its definition and scope are still evolving and expanding (Zhuang et al., 2023). With the development of society and technological progress, the content involved in architectural programming is constantly increasing, covering more emerging fields and technological applications. For example, with the popularization of sustainable development concepts, green building, and low-carbon design have become important considerations in architectural programming (Gibberd, 2005; Ristic Trajkovic et al., 2021). Additionally, the widespread application of digital technology has brought new challenges and opportunities to architectural programming, with technologies such as Information and Communication Technology (ICT), Building Information Modeling (BIM), Virtual Reality (VR), and Augmented Reality (AR) gradually changing the way architectural programming works and its processes (Barekati et al., 2015; Chen and Tang, 2019; Chung et al.,

2009; Shen et al., 2013). Faced with such trends, there is a need for a comprehensive analysis of architectural programming to delve into its challenges and development direction in a constantly changing social and technological context. At the same time, it is necessary to conduct comparative analyses of countries at different stages of development to provide more targeted recommendations.

This study focuses on a comparative analysis of architectural programming research and practice between China and other countries. This choice is based on the assumption that architectural programming developments and future directions in China and other countries share both commonalities and differences, and that their comparison can reveal valuable insights. On the one hand, China and other countries face similar challenges, such as accelerated urbanization, resource shortages and energy integration, increased pressures for sustainable development, and the impact of AI technologies (Wang et al., 2023; Yu et al., 2008). On the other hand, there are significant differences between China and other countries. Firstly, China, as one of the most populous countries in the world, has a rapidly developing construction industry (Wu et al., 2017; Zhang et al., 2014). However, compared to other countries, especially developed ones, there are distinct differences in the developmental stage, policy environment, and economic conditions of architectural programming (Hong et al., 2018; Liang et al., 2019). Secondly, the continuous investment by the Chinese government in urbanization and infrastructure construction has led to a growing demand for architectural programming (Deng and Poon, 2013; Shen et al., 2010). By comparing these commonalities and differences, we can better identify the shortcomings in China's architectural programming field, learn from the experiences and lessons of other countries, and further improve and enhance China's architectural programming standards. At the same time, this also provides valuable guidance for architectural programming in other countries, promoting the development and cooperation of the global construction industry.

The remaining structure of this study is as follows. Section 2 outlines the data collection process and the methods used in this study. Section 3 introduces the current research status in architectural programming, identifying annual publications, prominent contributors, keyword clusters, and research topics in this field. Section 4 discusses the research progress and hot topics in both China and other countries, comprehensively exploring the similarities and differences between China and other countries in the field of architectural programming. Finally, Section 5 summarizes the findings and conclusions of this study and provides directions for the future development of architectural programming.

2. Research methodology

To ensure the accuracy, transparency, and repeatability of the review, this study adopts a three-stage mixed research methodology. The core of this methodology includes bibliometric analysis, latent Dirichlet allocation (LDA) topic modeling, and systematic literature review, combining quantitative and qualitative analyses to comprehensively

evaluate the existing literature in the research field of architectural programming (Harden and Thomas, 2010).

Qualitative analysis serves as a crucial tool for facilitating an in-depth understanding of the literature and identifying gaps (McGowan and Sampson, 2005). Despite being susceptible to personal judgments and biases (Pearson et al., 2015), the use of bibliometric analysis as a quantitative and objective method can mitigate this limitation to a great extent (Heyvaert et al., 2016). Additionally, the text mining approach of LDA topic modeling enhances and optimizes bibliometric analysis, aiding in avoiding potential drawbacks and misunderstandings that may arise from solely relying on bibliometric analysis (Bilge and Yaman, 2022; Haustein and Larivière, 2015).

The three main stages of this research methodology are as follows: The first stage involves retrieving relevant literature from selected databases and screening them based on predefined criteria. In the second stage, the selected literature undergoes bibliometric analysis and LDA topic modeling to generate visual networks and quantitative data, providing an overview and clustering of the relevant literature. The third stage involves qualitative analysis of the screened literature to explore research progress and hot topics, and through comparative analysis of research differences between China and other countries, to summarize future research directions in architectural programming. The overall research framework is depicted in Fig. 1.

2.1. Data collection

The data for this study were sourced from the Web of Science (WoS) Core Collection and the China National Knowledge Infrastructure (CNKI) database. These databases cover a wide range of disciplines and fields, containing the world's most important and influential literature, providing reliable data support for this research (Web of Science Group, 2020; Zhang et al., 2020).

In WoS, the search terms "Architectural Programming" and "Briefing" were used, connected with the logical operator "OR". Considering that architectural programming research originated in the early 1960s (Cherry and Petronis, 2016), the search period was set from 1975 to 2024. As non-English literature constitutes a small proportion and LDA topic modeling cannot be performed across languages, "English" was selected as the document language. The document type was limited to "Article", which typically undergoes peer review and possesses higher academic quality and credibility. In CNKI, the search terms were "Architectural Programming" and "Architectural Planning", connected with the logical operator "OR". Considering that architectural programming was introduced in China after 1990 (Zhuang and Li, 1992), the search period was set from 1990 to 2024. "Journal articles" were selected as the document type. The search was conducted up to March 31, 2024, yielding 3404 articles from WoS and 558 articles from CNKI.

In the preliminary screening stage, articles that focused other than construction project types, described only project brief documents, or solely concentrated on design or construction phases were excluded. After screening, the

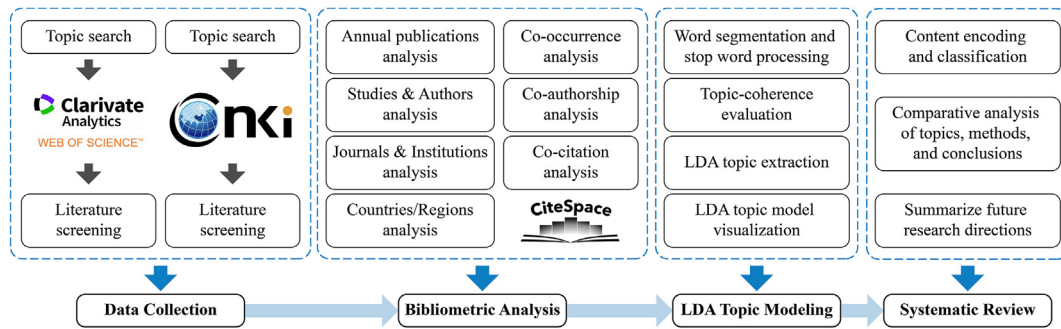


Fig. 1 Summary of the research methodology.

datasets from WoS and CNKI comprised 515 and 508 articles, respectively, for subsequent bibliometric analysis and LDA topic modeling. Furthermore, based on the titles, abstracts, and keywords of the articles, a qualitative screening was performed, resulting in 55 and 45 articles from WoS and CNKI, respectively, eligible for systematic literature review.

2.2. Bibliometric analysis

In this study, the widely-used literature analysis tool, CiteSpace, was employed for bibliometric analysis. This tool possesses the capability for large-scale literature analysis and visualization, including co-occurrence analysis, co-citation analysis, and co-authorship analysis, enabling a better understanding of the evolutionary path and core nodes in the research field (Chen, 2017; Chen et al., 2009). Additionally, its flexible data management function allows for better integration with other data processing platforms (Chen, 2016).

Firstly, co-occurrence analysis was utilized to explore keyword clusters in the research field, aiding in the formation of clear research topics or research groups. This facilitates the clarification of the intrinsic relationships within the research field and identifies the interactions and intersections among different research topics (Thelwall, 2008). Secondly, co-citation analysis was selected to identify the main research progress in the field. By analyzing nodes or clusters with sudden increases in citation frequency, emerging hotspots or important development trends in the research field were captured, providing insights into the latest research trends and focuses (He et al., 2017).

2.3. LDA topic detection

Although bibliometric analysis excels in quantitative metrics and data, it often overlooks the deeper information and multidimensional correlations within the literature, and is prone to citation biases, leading to the underestimation or neglect of some important studies (Choi and Seo, 2020). Therefore, this study employs LDA topic modeling, a text-mining method, to optimize and enhance the effectiveness of bibliometric analysis (Jelodar et al., 2019).

Text mining is a process of automatically discovering potential, previously unknown, and valuable information

from large-scale text data (Ezzeldin and El-Dakhakhni, 2020). Within text mining, LDA topic modeling is one of the important methods, characterized by its high flexibility and effectiveness. It can effectively uncover latent topics hidden within text data, providing researchers with more accurate and reliable research conclusions (Blei et al., 2003). The model assumes that each document consists of multiple topics, and each topic is composed of multiple words. By analyzing the distribution of words in documents and the distribution of topics in words, the topic distribution of documents can be inferred. In this study, LDA topic modeling was applied to further explore the clustering results of co-citation analysis generated by CiteSpace, ultimately determining the topics in each trend within the research field. Before conducting LDA topic detection, text preprocessing was conducted, including noise removal, establishing and filtering stop-word lists, and creating and applying fixed vocabularies. Subsequently, the results of LDA topic detection were merged with the clustering labels from CiteSpace to obtain deeper and more comprehensive topic structure information for each research topic.

2.4. Systematic review

The systematic literature review aims to conduct an in-depth qualitative analysis of the selected literature to compare research disparities between China and other countries and to summarize future research directions in this field (Favras et al., 2022).

Initially, a thorough reading and comprehension of the selected literature are undertaken, with its main contents being coded and classified (Peiris et al., 2023). Emphasis is placed on aspects such as research methods, theoretical frameworks, research findings, research conclusions, and future research directions (McGowan and Sampson, 2005). Through comprehensive literature analysis, a holistic understanding of the research progress and development trends in the field of architectural programming can be achieved. Subsequently, particular attention is paid to the similarities and differences between Chinese and other countries' research. Through comparative analysis, disparities between Chinese and other countries' literature on research topics, methods, and conclusions are identified. This process aims to unearth the research characteristics and advantages of scholars in the field of architectural programming, providing references and insights for subsequent research. Finally, future research directions in

architectural programming are summarized. Based on the results of the comparative analysis, targeted and practically meaningful future research directions are proposed. These directions will provide theoretical support and practical guidance for academic research in the field of architectural programming.

3. Overview and cluster results

3.1. Annual publications

The annual publication quantities from the WoS and CNKI databases were analyzed separately using bibliometric tools, with the results shown in Fig. 2.

From 1975 to 2024, there has been a steady growth trend in the annual publications in the field of architectural programming in the WoS database. Initially, between 1975 and 1990, there were relatively few publications, with only a small number of articles published. Subsequently, from 1991 to 2000, there was a slight increase in publication quantity, but the overall growth was slow. Starting from 2001, there was a significant increase in the number of publications, showing a clear upward trend. Particularly after 2018, the annual publication quantity has consistently remained above 30 articles.

In contrast, the annual publication quantity in the field of architectural programming in the CNKI database has shown more unstable variations. Between 1990 and 2005, there was a relatively slow growth in publication quantity, especially during the period from 1990 to 2000, when there were few publications. Starting from 2005, the number of publications gradually increased, peaking between 2015 and 2019. During this period, there was considerable attention and development in the field of architectural programming research. However, from 2015 to 2024, there has been a downward trend in publication quantity, particularly accelerating after 2021.

3.2. Prominent contributors

Identifying and summarizing prominent contributors is crucial when examining the current status and development trends of a field. Their research achievements not only reflect the research hotspots and trends in the field but also guide future research directions and collaborations. This study identified five types of prominent contributors: documents, authors, journals, institutions/organizations, and countries/regions, and compiled relevant data from the WoS and CNKI databases.

Firstly, Tables 1 and 2 respectively list important research related to architectural programming in the WoS and CNKI databases. Among them, the top 10 most cited articles in the WoS database mainly focus on important topics such as architectural cost prediction and optimization, critical success factors analysis, and user pre-assessment methods. In contrast, the most cited articles in the CNKI database primarily cover key areas such as programming theory and methods, post-occupancy evaluation, and building lifecycle management.

Secondly, Tables 3 and 4 summarize the most influential authors in the field of architectural programming in the WoS and CNKI databases, respectively. In the WoS database, the most cited authors include Leaman, A., Shen, QP, and Yu, ATW, whose research areas encompass architectural performance evaluation, public-private partnership (PPP) construction projects, and early-stage construction management. In the CNKI database, the most cited authors include Zhuang, W., Zhang, W., and Tu, H., whose research mainly focuses on architectural spatial target assessment, full-process post-occupancy evaluation, and multi-agent group decision theory.

Thirdly, Tables 5 and 6 respectively summarize the most cited journals in the field of architectural programming in the WoS and CNKI databases. In the WoS database, the most cited journals include “Building Research and Information,” “Automation in Construction,” and “Journal of Construction

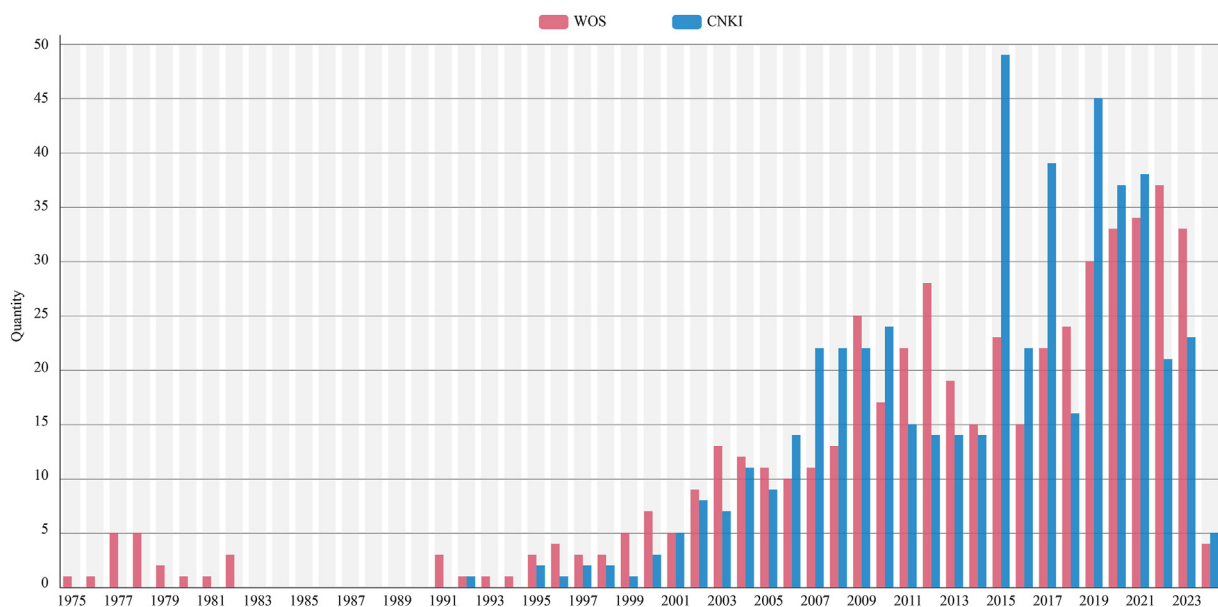


Fig. 2 Annual publication quantity statistics from WoS and CNKI databases.

Table 1 Top 10 most highly-cited studies on architectural programming from WoS.

No.	Title	Author	Journal	Citation Count	Year
1	Prediction and optimization of life-cycle costs in early design	Bogenstätter, U	Building Research and Information	127	2000
2	Ranked Critical Factors in PPP Briefings	Tang, LY	Journal of Management in Engineering	78	2013
3	Investigation of critical success factors in construction project briefing by way of content analysis	Yu, ATW	Journal of Construction Engineering and Management	77	2006
4	The User Pre-Occupancy Evaluation Method in designer-client Communication in Early Design Stage: A Case Study	Shen, WL	Automation in Construction	46	2013
5	The design brief as carrier of client information during the construction process	Ryd, N	Design Studies	46	2004
6	Architect and user interaction: the spoken representation of form and functional meaning in early design conversations	Luck, R	Design Studies	35	2006
7	Improving megaproject briefing through enhanced collaboration with ICT	Chung, JKH	Automation in Construction	30	2009
8	Capturing the social value of buildings: The promise of Social Return on Investment (SROI)	Watson, KJ	Building and Environment	25	2016
9	Design quality in pre-project planning: applications of the Project Definition Rating Index	Gibson, GE	Building Research and Information	24	2003
10	Demystifying Moderate Variables of the Interrelationships among Affective Commitment, Job Performance, and Job Satisfaction of Construction Professionals	Leung, MY	Journal of Construction Engineering and Management	23	2008

Engineering and Management.” Among them, “Building Research and Information” demonstrates its significant influence in the field through its high citation frequency, article quantity, and strong connections. Although “Automation in Construction” and “Journal of Construction Engineering and Management” have relatively fewer articles, their high citation frequency and strong connections also indicate their research quality and impact. In contrast, the most cited journals in the CNKI database include “Architectural Journal,” “Design Community,” and “New Architecture.” Among them, “Architectural Journal” showcases its remarkable influence in the field through its highest average citation frequency. Although “Design Community” and “New Architecture” have slightly fewer citations and articles, their relatively high citation/article ratio reflects their recognition and importance in the academic community.

Fourthly, [Tables 7 and 8](#) respectively summarize the institutions that have published the most research related to architectural programming in the WoS and CNKI databases. The institutions with the highest number of publications in the WoS database include Hong Kong Polytechnic University, Delft University of Technology, and Loughborough University. Among them, the papers from Hong Kong Polytechnic University are widely cited,

indicating its independent position in the international academic community. While institutions like Delft University of Technology and Loughborough University slightly lag in citation counts, they also wield considerable influence in the field. It’s worth noting that the collaboration and communication among these influential institutions are relatively low, suggesting a preference for independence in research and practice. In contrast, the institutions with the most publications in the CNKI database include Tsinghua University, Tongji University, and the Architectural Design and Research Institute of Tsinghua University. Tsinghua University stands out with 2231 citations and 72 articles, showcasing its significant position and influence in the field of architectural programming in China. While institutions like Tongji University and Harbin Institute of Technology have fewer citations and articles, they still contribute to research in this field.

Lastly, [Table 9](#) summarizes the countries with the most research related to architectural programming published in the WoS database, including England, the Netherlands, the USA, China, and Australia. Among these countries, England leads with 1193 citations and 150 articles, demonstrating its significant influence and recognition in the field. While countries like the Netherlands and the USA have slightly

Table 2 Top 10 most highly-cited studies on architectural programming from CNKI.

No.	Title	Author	Journal	Citation Count	Year
1	SD method related to evaluation of architectural space objectives	Zhuang, W	Journal of Tsinghua University	588	1996
2	Reviewing the old and learning the new: Introduction to Post Occupancy Evaluation (POE) method	Han, J	Architectural Journal	350	2006
3	20 Application Types of BIM in a Building's Lifecycle	Guo, J	Architecture Technique	264	2011
4	Post-Occupancy Evaluation: A Review of Methods and State-of-the-Art Techniques	Zhuang, W	Time + Architecture	98	2019
5	Architectural programming theory: Exploration of design methodology	Zhuang, W	Architectural Journal	95	1992
6	From architectural programming to architectural design	Zhuang, W	New Architecture	86	1997
7	From Architectural Programming to Post Occupancy Evaluation: A Feedback Mechanism of Architectural Procedure Circle	Zhuang, W	Design Community	80	2017
8	On the Method of Spatial Prediction and Evaluation in Architectural Programming: Feedforward of Post Occupation Evaluation	Su, S	New Architecture	68	2011
9	Designing for the Countryside: The Architectural Programming Method as a Strategy	Dang, T	Architectural Journal	62	2019
10	POE: the Criteria of a Reasonable Design	Zhuang, W	Design Community	53	2017

fewer citations, they still wield considerable influence. Additionally, although China started its research in architectural programming relatively late compared to other countries, it has shown impressive performance with increasing citations and publications, as well as relatively high connection strength. Meanwhile, countries like Australia and Germany, while having fewer citations and publications, still attract attention in this field.

To further analyze the research dynamics and trends in architectural programming between China and other countries, this study conducted a selection of all literature from China in the WoS database and combined it with the CNKI database to form the Chinese database. The remaining WoS database was treated as the database for other countries, providing more comprehensive data support and analysis basis for subsequent analysis.

Table 3 Top 5 most highly-cited authors and their representative works on architectural programming from WoS.

No.	Most Influential Author	Citations (All Scopes)	Masterpiece	Source
1	Leaman, A	175	Assessing building performance in use 5: conclusions and implications	Building Research and Information
2	Shen, QP	171	Ranked Critical Factors in PPP Briefings	Journal of Management in Engineering
3	Yu, ATW	112	Comparative study of the variables in construction project briefing/architectural programming	Journal of Construction Engineering and Management
4	Tang, LY	96	Procurement-Related Critical Factors for Briefing in Public-Private Partnership Projects: Case of Hong Kong	Journal of Management in Engineering
5	Gibson, GE	81	Preproject-Planning Process for Capital Facilities	Journal of Construction Engineering and Management

Table 4 Top 5 most highly-cited authors and their representative works on architectural programming from CNKI.

No.	Most Influential Author	Quantity	Masterpiece	Source
1	Zhuang, W	40	SD method related to evaluation of architectural space objectives	Journal of Tsinghua University
2	Zhang, W	19	A Review of Research Progress on Post-Occupation Evaluation Based on the Closed-Loop System of "Analysis-first-Evaluation-after"	Time + Architecture
3	Tu, H	15	A Study on Group Decision Makers of Architectural Programming of Large Complex Projects	Architectural Journal
4	Zhou, G	10	Community-Based Comprehensive Old-Age Service Facilities Design based on Social Design Theory	Architectural Journal
5	Qu, Z	9	Application of Architectural Programming Based on Multi-subject in Urban Renewal	Design Community

Table 5 Top 5 most highly-cited journals on architectural programming from WoS.

No.	Journal Title	Citations	Documents	Total link strength
1	Building Research and Information	399	11	59
2	Automation in Construction	182	5	72
3	Journal of Construction Engineering and Management	117	3	115
4	Journal of Management in Engineering	115	4	96
5	Engineering Construction and Architectural Management	36	4	51

3.3. Keywords analysis

Keyword analysis of literature in the field of architectural programming in China and other countries was conducted to reveal the connections and influences among studies, thereby gaining a deeper understanding of the structure and development trends in this field.

Firstly, co-occurrence analysis of keywords in literature from other countries was conducted using CiteSpace, as shown in Fig. 3. Keywords reflect core concepts and topics in the literature, with the size of nodes and fonts proportional to their frequency of occurrence in the literature. In

addition to "architectural programming" and "briefing" as core terms, common keywords include "education," "social impact," "design," "performance," "model," "impact," "environment," "construction," "mathematical modeling," and "behavior." Furthermore, keywords with higher centrality include "education," "design," "model,"

Table 6 Top 5 most highly-cited journals on architectural programming from CNKI.

No.	Journal Title	Citations	Documents	Citations/ Documents
1	Architectural Journal	867	27	32.11
2	Design Community	354	53	6.68
3	New Architecture	335	25	13.4
4	Huazhong Architecture	208	30	6.93
5	Urbanism and Architecture	149	35	4.26

Table 7 Top 8 institutions/organizations for frequently published studies on architectural programming from WoS.

No.	Research Institute/organization	Citations	Documents	Total link strength
1	Hong Kong Polytechnic University	303	10	1
2	Delft University of Technology	152	5	0
3	Loughborough University	129	9	0
4	University of Antwerp	126	6	0
5	University of Oxford	90	8	1
6	University College London	78	11	1
7	Trinity College Dublin	76	8	0
8	University of Manchester	50	7	1

Table 8 Top 8 institutions/organizations for frequently published studies on architectural programming from CNKI.

No.	Research Institute/organization	Citations	Documents	Total link strength
1	Tsinghua University	2231	72	30
2	Tongji University	359	43	2
3	Architectural Design & Research Institute of Tsinghua University	173	19	27
4	Harbin Institute of Technology	123	15	0
5	Nanjing University of Technology	15	14	1
6	Southeast University	63	12	1
7	Chongqing University	61	12	0
8	Xi'an University of Architecture and Technology	39	12	2

“behavior,” and “environment,” indicating their significance in the overall network connectivity and influence.

Secondly, a co-occurrence analysis of keywords in literature from China was conducted using CiteSpace, as shown in Fig. 4. In addition to the core term “architectural programming,” common keywords include “architectural design,” “urban renewal,” “SD method,” “post-evaluation,” “sports architecture,” “architects,” “green buildings,” “assignment book,” “space,” and “multi-agent.” Moreover, keywords with higher centrality include “architectural design,” “hospital,” “value assessment,” “architects,” and “green buildings,” indicating their importance in the overall network connectivity and influence.

A co-citation analysis of keywords in literature from other countries was conducted, and clustering was performed based on the co-citation relationships among the literature. As shown in Fig. 5, 515 articles were classified into 7 co-citation clusters using CiteSpace. Each cluster represents a group of literature with a high correlation in co-cited keywords. In the figure, each row represents the same topic cited by different literature, with the size of connecting nodes indicating the frequency of citation of that literature and the connecting lines representing co-citation among different literature. Overall, the distribution of research topics in architectural programming in other countries is relatively uniform, covering multiple relatively independent key topics. The clusters are arranged in descending order based on the number of literature, where Cluster #0 involves “client requirement,” Cluster #2 focuses on “environment,” Cluster #3 involves “mathematical modeling,” and so on. It is noteworthy that

the labels of each cluster are noun phrases selected from the titles, keywords, and abstracts of the literature in the cluster. These noun phrases are derived using the Log-Likelihood Ratio (LLR) clustering label extraction algorithm in CiteSpace.

A co-citation analysis of keywords in literature from China was conducted, and clustering was performed based on the co-citation relationships among the literature. As shown in Fig. 6, 508 articles were classified into 20 co-citation clusters using CiteSpace. Overall, China has a very concentrated research source in the field of architectural programming, which has extended to multiple different research topics. Although there are certain differences among the various research topics, they are closely associated with the research source, indicating the continuity and depth of research in the field of architectural programming in China. The clusters are arranged in descending order based on the number of literature, where Cluster #0 involves “architectural design,” Cluster #2 focuses on “assignment book,” Cluster #3 involves “architecture,” and so on.

Table 10 lists the basic information of each co-citation cluster for both China and other countries. The silhouette score is an indicator to evaluate the quality of clustering results, usually inversely related to the size of the cluster. Its value ranges from −1 to 1, where 1 indicates that the cluster label completely reflects the topic and research trends of the literature under that cluster (Chen, 2016). In this study, the silhouette score of the 7 co-citation clusters from literature in other countries ranges from 0.934 to 0.984, and the silhouette score of the 20 co-citation

Table 9 Top 10 countries/regions for frequently published studies on architectural programming from WoS.

No.	Country/Region	Citations	Documents	Total link strength
1	England	1193	150	21
2	Netherlands	1125	26	15
3	USA	723	52	23
4	China	489	26	27
5	Australia	488	31	33
6	Germany	201	17	13
7	Belgium	196	10	3
8	Scotland	160	16	9
9	Denmark	35	2	5
10	Malaysia	34	9	3



Fig. 4 Co-occurrence network of keywords in the literature from China.

due to the possibility of the co-citation network models containing the same reference literature topics. Additionally, the year distribution of clusters reveals research trends, but this does not imply that these topics are not studied outside the trend, as research may involve multiple topics, and the year distribution only reflects the focus of research in different periods.

4. Research progress

Research topics in architectural programming in China can be summarized in 8 aspects, including programming theory and methods, intelligent design and technology, infrastructure construction, urban planning and rural revitalization, sustainable development and environmental protection, architectural education and talent cultivation, project management and decision analysis, as well as social participation and multi-stakeholder cooperation. In recent years, research on architectural programming in China has shown a trend toward diversification and deepening (Zhuang et al., 2023). For instance, there has been a deeper exploration into topics such as building assignment books, preprogramming, and post-evaluation to meet the evolving needs of urban construction (Liu, 2020; Miao et al., 2020). Furthermore, with the continuous development and popularization of BIM technology, researchers in China have

gradually applied BIM and other digital technologies to architectural programming to improve the efficiency and quality of architectural design (Lv, 2022).

In contrast, research topics in architectural programming in other countries can be summarized in 6 aspects, including project management and construction practice, digital technology and collaborative work, sustainable architectural design and practice, architectural education and professional development, urban renewal and planning decisions, as well as public service facility construction and management. Research in these countries covers many cutting-edge areas such as the application of intelligent technology and innovative collaborative practices (Gu et al., 2023; Khosrowshahi, 2015). Additionally, there is a certain emphasis on research into sustainable development and green architecture, reflecting a high level of concern for environmental protection and sustainable development (Heenan, 2013; Zhang et al., 2017).

In summary, with the advancement and interdisciplinary integration of the discipline, along with the diversification and complexity of construction projects, the theories and methods of architectural programming have gradually extended to encompass project management, urban design, and higher education. Consequently, current research not only focuses on identifying and analyzing various decision-making criteria but also explores how to optimize architectural programming theories and practices

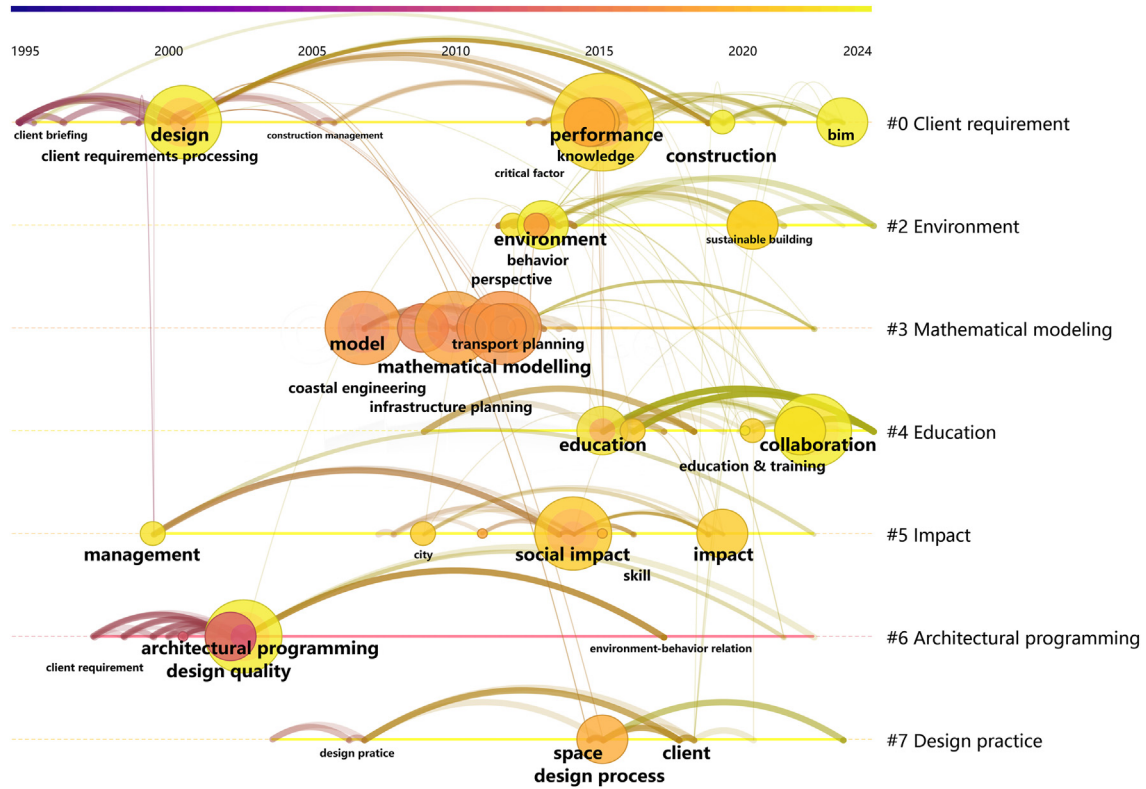


Fig. 5 Co-citation network of keywords in the literature from other countries.

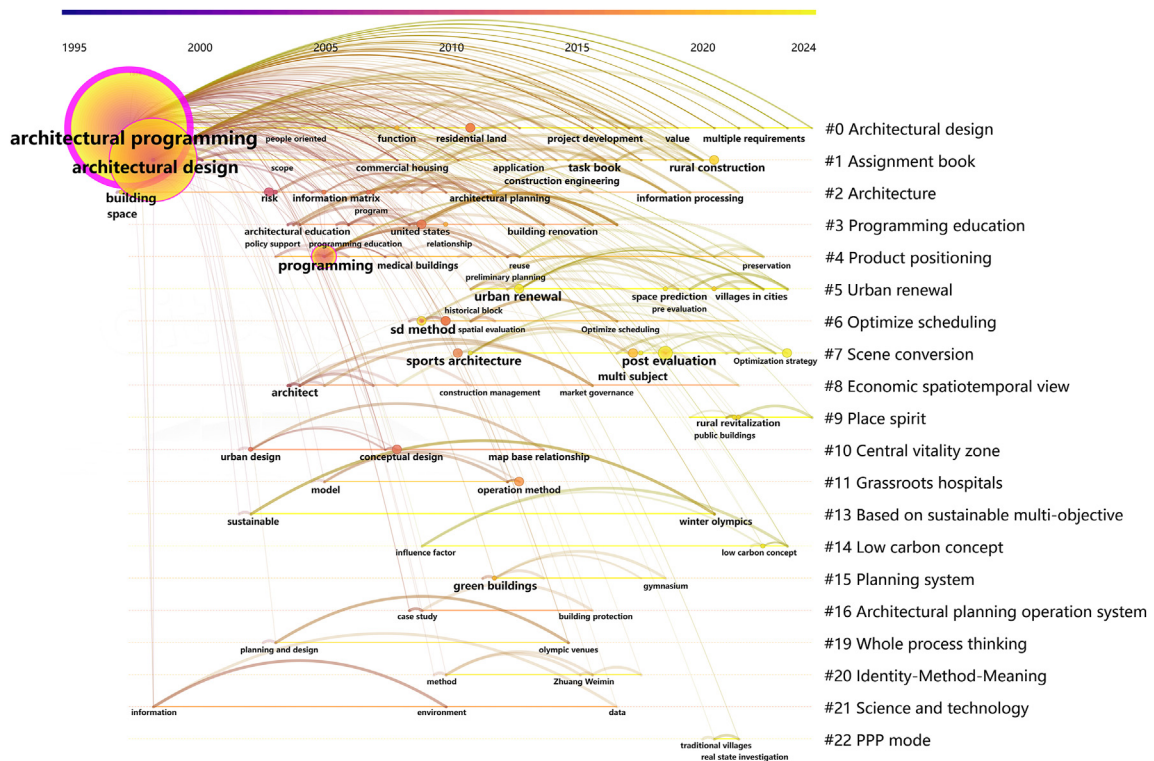


Fig. 6 Co-citation network of keywords in the literature from China.

Table 10 Information statistics of each co-citation cluster.

Category	Cluster-ID	Size	Silhouette	mean (Year)	Top Terms (log-likelihood ratio, <i>p</i> -level)
Other countries	0	62	0.954	2011	Client requirement
	2	37	0.981	2016	Environment
	3	37	0.934	2012	Mathematical modeling
	4	36	0.946	2020	Education
	5	28	0.947	2015	Impact
	6	24	0.984	2008	Architectural programming
	7	22	0.97	2015	Design practice
China	0	137	0.986	2014	Architectural design
	1	65	0.897	2011	Assignment book
	2	37	0.893	2008	Architecture
	3	28	0.934	2008	Programming education
	4	27	0.959	2010	Product positioning
	5	21	0.981	2019	Urban renewal
	6	14	0.954	2012	Optimize scheduling
	7	14	0.959	2017	Scene conversion
	8	13	0.968	2007	Economic spatiotemporal view
	9	9	0.993	2021	Place spirit
	10	9	0.966	2007	Central vitality zone
	11	8	0.99	2010	Grassroots hospitals
	13	6	0.994	2013	Based on sustainable multi-objective
	14	6	0.976	2020	Low carbon concept
	15	6	0.986	2013	Planning system
	16	5	0.996	2009	Architectural planning operation system
	19	5	0.994	2009	Whole process thinking
	20	5	0.995	2013	Identity-Method-Meaning
	21	5	0.998	2008	Science and technology
	22	4	0.998	2020	PPP mode

through systematic and multidisciplinary decision-making processes. To comprehensively compare and analyze the research progress in architectural programming between China and other countries, this study summarizes the following six research areas: intelligent and digital methods, sustainable development and practice, project management and decision support, urban planning and existing development, social infrastructure and public services, and architectural education and talent cultivation.

4.1. Intelligent and digital methods

In the field of architectural programming, the application of intelligent and digital methods is increasingly valued as an important means to enhance architectural design efficiency and optimize project management. Both China and other countries have their focuses in this area but are striving to explore paths of technological innovation and application.

Research on intelligent and digital methods in architectural programming in China mainly focuses on BIM practice, big data analytics, and machine learning, among others. [Qu \(2019\)](#) proposed a pre-assessment method for architectural programming based on BIM collaborative analysis technology, which conceptualized three-dimensional and four-dimensional pre-assessment methods to enhance the scientificity of architectural programming at the early stage. [Liu \(2019\)](#) utilized data

mining methods to propose an analysis method for area table data based on a briefing sample library, providing architects with a data-based inspection to quickly identify potential problems in area proportions. [Chang et al. \(2015\)](#) proposed a Data Science-Assisted Design (DSAD) method, applying informatics and systems theory in design-related fields to construct a widely applicable work platform, achieving functions such as data recording, analysis, evaluation, and prediction.

In other countries, research on intelligent and digital methods mainly focuses on agile development frameworks, supervised machine learning, fuzzy hybrid simulation, and other aspects. [Jelicic et al. \(2023\)](#) proposed an architectural programming and project management method based on Scrum, a well-known agile development framework, significantly shortening the design process and facilitating the participation of more experts in the same project team. [Platt et al., 2024](#) used supervised machine learning methods to analyze the relationship between regional exterior design factors of healthcare systems and Hospital-Acquired Infections (HAI), providing decision support for the programming and pre-design of hospital buildings. [Vahabi et al. \(2022\)](#) proposed a fuzzy hybrid simulation method, simulating the impact of briefing clarity on project cost and time performance, and providing an effective approach for the assessment of cost and time performance in the early stages of project delivery.

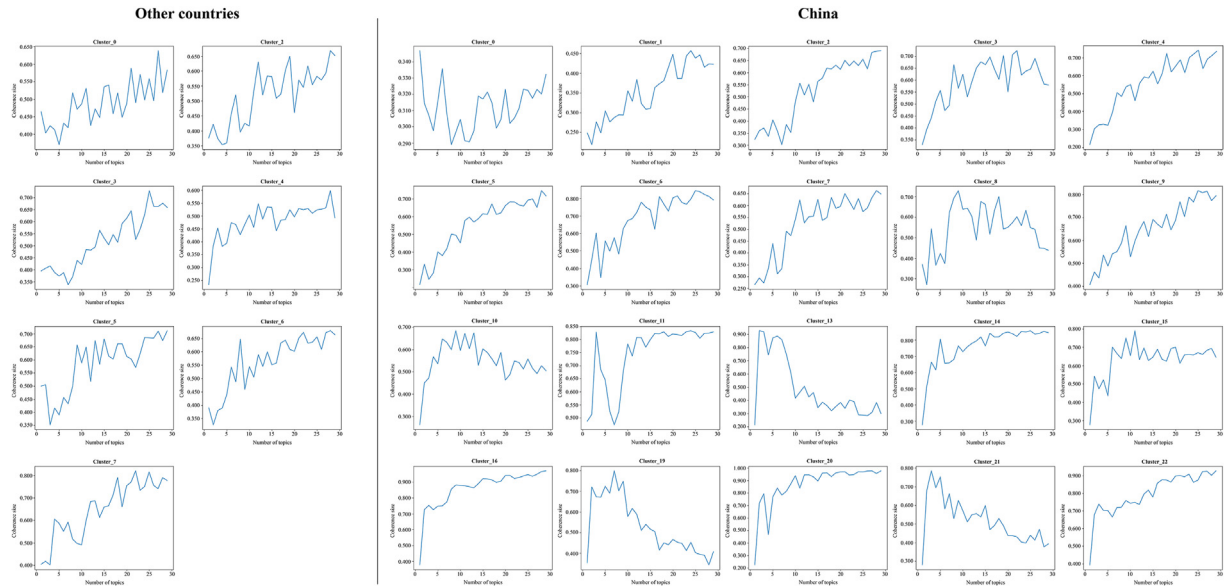


Fig. 7 The topic-coherence score curve of co-citation clusters between China and other countries.

In summary, both China and other countries are actively exploring the application of various intelligent and digital methods, such as AI, ICT, unmanned aerial vehicles (UAVs), VR (Chung et al., 2009; Khosrowshahi, 2015; Shen et al., 2013; Dang et al., 2020). These technological tools are widely used in different stages of architectural programming to improve efficiency, reduce costs, and optimize design schemes. On the one hand, research in China on intelligent and digital methods tends to focus more on the application and development of technology in architectural practice, aiming to improve design efficiency and quality (Liu and Zhuang, 2016). However, other countries tend to focus more on the application of technology in

design innovation and management efficiency, striving to enhance the flexibility of architectural design and the controllability of project management (Vahabi et al., 2023). On the other hand, research in China mainly focuses on exploring the application of existing tools in the field of engineering technology, such as BIM-based pre-assessment methods and big data analytics methods. In contrast, other countries adopt interdisciplinary methods such as agile development frameworks, supervised machine learning, and fuzzy hybrid simulation, taking a more macroscopic and systematic perspective to address architectural programming issues and proposing innovative research methods and ideas.



Fig. 8 The distribution of topic models of co-citation clusters between China and other countries.

Table 11 Research topics and year trends of co-citation clusters between China and other countries.

Category	Cluster-ID	Research topics	Research trends
Other countries	#0	Construction safety management Debriefing in project management Factors influencing project success Energy-efficient construction practices Simulation and digital technologies in construction Collaborative project management	1995–2023
	#2	Digital collaborative workspaces Heritage conservation and architectural design Environmental design and engineering Community engagement and design Hybrid learning models in architectural education	2012–2024
	#3	Infrastructure design and management Knowledge sharing and best practices Engineering practice and professional development Infrastructure standards and directives	2007–2022
	#4	Architectural education and collaborative learning Assessment and feedback in architectural practices Interprofessional collaboration in architectural design Simulation-based learning	2009–2024
	#5	Policy decision-making in urban development Client requirements and project briefings Modeling urban systems and territorial planning Decision support technologies (DST) for urban planning	2000–2022
	#6	Urban design and planning for heritage preservation Housing Planning and Social Values Future-oriented urban design frameworks Quality-based programming for architectural projects	1998–2022
	#7	Public service design and management User-centered design in public spaces Data-driven decision-making in construction Legal and constitutional aspects of public design	2004–2023
	#0	Architectural programming “Analysis-first-Evaluation-after” Architect responsibility system Real estate projects Architectural space evaluation	1996–2024
	#1	Architectural design Building Assignment Intelligent design Hospital architectural design BIM	1997–2020
	#2	Post occupancy evaluation Historical environmental assessment SD method Risk analysis	1996–2021
	#3	Architecture teaching Professional education for architects Current situation of programming education development Programming education in the United States	2003–2016
	#4	Full process consultation Project development and value planning Industrial building programming Medical building programming	2003–2022
	#5	Renovation of old residential areas	2010–2023
			(continued on next page)
China			

Table 11 (continued)

Category	Cluster-ID	Research topics	Research trends
		15-min life circle	
		Historical block renewal	
		Multi-subject participation	
	#6	Classroom environment evaluation	2008–2021
		Campus planning and design	
	#7	Mega project decision-making	2009–2023
		Comprehensive service facilities	
		Sports architecture programming	
	#8	Architects	2003–2021
		Owners/clients	
	#9	Rural construction	2019–2024
		Rural revitalization	
	#10	Urban design	2001–2013
		Urban map-based relationship	
	#11	Operation mode and method	2004–2012
	#13	Sustainable multi-objective	2001–2019
	#14	Project influencing factors	2008–2023
		Low carbon concept	
	#15	Green building design and technology	2010–2017
		Environmental characteristic analysis	
	#16	Operating system and case studies	2007–2014
	#19	Planning design	2001–2013
	#20	Programming methods	2009–2016
	#21	Information model	1997–2015
	#22	Protection of traditional villages	2019–2021

4.2. Sustainable development and practice

Emphasizing and applying the concept of sustainable development in architectural programming is crucial for enhancing the environmental friendliness, resource utilization efficiency, and social benefits of architectural design. Both China and other countries have their focuses in this field but are committed to promoting sustainability in various stages of the architectural industry, including design, construction, operation, and dismantlement, throughout the entire lifecycle (Andreu and Oreszczyn, 2004; Qian and Cheng, 2017).

Research on sustainable development and practice in the field of architectural programming in China mainly focuses on the analysis of architectural micro-environmental characteristics, simulation of green building performance, and sustainable building operation. Li et al. (2017) achieved comfortable microenvironments and site stormwater management in the design of a sports and cultural center through climate feature analysis, such as wind environment and solar radiation, providing important experience for green building design. Liu (2015) emphasized the key role of digital simulation technology in green building programming and proposed the importance of digital simulation technology in the refinement design process, providing a reliable tool for achieving green buildings. Zhang et al. (2020) proposed a multi-objective integrated programming approach that focuses on competitive performance, observation experience, and organizational efficiency by studying the sustainable integration of winter biathlon

venue programming, providing references for the construction of Winter Olympics National Biathlon Centers.

In other countries, research on sustainable development and practice focuses more on improving building energy efficiency, low environmental impact building design, and environmental and microclimate responsiveness. Berawi et al. (2023) proposed an intelligent office space design framework, which improved building energy performance and quality by integrating IoT technology into the value engineering process. Zhang et al. (2017) proposed design strategies for low environmental impact housing in tropical regions through the fuzzy Delphi method and analytic hierarchy process, emphasizing the importance of architectural programming and passive design in the widespread acceptance of green housing. Heenan (2013) showcased how buildings in four Australian sports stadiums achieve sustainability based on local topography and microclimate conditions, as well as education on sustainable buildings.

In summary, research in China and other countries on sustainable development and practice overlaps to a certain extent, focusing on the surrounding environment of buildings, green building performance, and so on. Both China and other countries are aware of the impact of the construction industry on the environment and resources and are seeking to integrate the concept of sustainable development into architectural programming to reduce negative environmental impacts and improve the efficiency and quality of architectural projects. However, research in China in this field tends to focus more on the sustainability and actual effects of building operations, while other

countries place more emphasis on improving building energy efficiency and reducing environmental impacts. These differences also reflect the different stages of development in the field of architectural programming in each country. China is in the process of comprehensive development of the construction industry, so more emphasis is placed on the practical process and the practical effects of sustainable intervention. Other countries have more mature research in these areas, thus focusing more on building energy efficiency and site environmental impacts, exploring new directions for the sustainable development of the global construction industry.

4.3. Project management and decision support

The research field of architectural programming often involves the aspect of project management and decision science. The introduction of this aspect not only ensures the smooth progress and high-quality completion of architectural projects but also provides effective support and guidance in the decision-making process, thereby achieving the best results for the project within the set objectives (Kelly and Duerk, 2002; Zhuang, 2017).

In the field of architectural programming, research in China on project management and decision support mainly includes architect responsibility systems, whole-process consultancy, and large-scale project decision-making. Huang et al. (2021) investigated the successful factors of the architect responsibility system in the renovation project of Eataly in Turin, Italy. Through the architect's leading involvement, multi-level updates of planning, architecture, interior, and landscape were achieved, enhancing the urban environmental quality. Zhang (2021) reviewed the international background and development trends of whole-process engineering consultancy, the relationship between architectural programming and whole-process engineering consultancy, and analyzed regulations, standards, scientific research practices, and talent cultivation related to architectural programming, and finally made suggestions for the future development of whole-process engineering consultancy in China. Tu and Su (2016) explored the introduction of group decision-making in the architectural programming of large complex projects based on architectural programming and group decision-making theories. They discussed in detail issues such as decision-making subjects, composition, and weight allocation, providing important references for project decision-making.

In other countries, research in the field of architectural programming on project management and decision support focuses more on critical success factors of projects, value management and risk identification, and establishment of decision support systems. Yu and Shen (2015) systematically identified critical success factors in the architectural programming process through questionnaire surveys and factor analysis, providing guidance for the success of architectural projects. Lee et al. (2023) explored the application of value management in architectural programming and proposed guidelines for childcare center programming through multi-party participation and communication, providing the scientific basis for construction project decisions. In addition, Lee et al. (2022a)

optimized the decision-making process of design-build (D-B) projects by developing decision support tools, improving the efficiency and accuracy of the decision-making process. The decision-making tool can also positively influence novice decision-makers.

In summary, research in China and other countries on project management and decision support has similarities, as they both aim to improve the management efficiency and decision quality of architectural projects. Specifically, they both focus on exploring how to optimize project management processes and improve the success rate of project implementation. For example, both emphasize the analysis of factors of projects to identify critical factors affecting project success and take corresponding measures to manage and optimize them. Additionally, they both discuss the establishment and application of decision support systems to assist decision-makers in making scientific and rational decisions, thereby promoting the smooth progress of projects. However, research in China tends to emphasize the leading role of architects and their full participation throughout the project to ensure smooth implementation and quality assurance. Research in other countries, on the other hand, focuses more on the cooperation, participation, and joint decision-making among the various stakeholders of the project, and emphasizes the improvement of the level of project management and decision-making through scientific analysis and the application of tools. Furthermore, there are differences in research methods and theoretical frameworks, with research in China leaning towards empirical analysis and case studies, while research in other countries focuses more on theoretical models and quantitative analysis.

4.4. Urban planning and existing development

Urban planning and existing development impact various aspects of urban spatial structure, functional layout, transportation, resource utilization, and environmental protection. Scientific and rational programming and construction can improve land use efficiency, optimize urban and rural development layouts, and promote economic development and social progress.

In China, research on urban planning and existing development mainly focuses on historical district renewal, renovation of existing urban buildings, and rural revitalization. Liu et al. (2019), using the comprehensive renewal design of Yuliang South Road in Jiujiang City as an example, proposed strategies for historical district renewal based on architectural programming theories and methods. They put forward corresponding transformation strategies for buildings of different levels and attributes, providing a scientific basis for historical district renewal. Huang et al. (2022), using the renovation project of Fermi High School in Turin as an example, discussed the programming and renovation methods of existing buildings in urban renewal, emphasizing the importance of reasonable time allocation and close cooperation among multiple stakeholders for the success of the project. Zou and Qu (2021) explored the "planning-programming-architecture" whole-process design method in rural elderly care facility planning, aiming to achieve effective integration of rural construction

resources in terms of function, form, and operation, providing new ideas for rural revitalization.

In other countries, research on urban planning and existing development focuses more on long-term urban visions and planning, urban heritage conservation, and activation of historical buildings. Rogers and Hunt (2019) proposed a future-oriented urban vision method to guide urban programming and development by constructing long-term visions, emphasizing the multiple benefits of engineering interventions. Kalayci and Ozdemir (2021), using Victory Square in Ankara, Turkey, as an example, proposed urban planning and programming strategies to inherit historical value through historical investigation and on-site analysis, aiming to pass on past values to the future. Pyburn (2017) discussed the application of agreement-based architectural programming methods in the renovation of historical buildings, emphasizing the importance of preserving historical features and creatively connecting the past with the future, resulting in unique design outcomes that preserve historical characteristics in the activation of historical buildings.

In conclusion, China and other countries have different focuses in research on urban planning and existing development, but both recognize the importance of urban historical culture for social development and therefore attach importance to the protection and activation of historical buildings. Additionally, both China and other countries regard sustainable development as an important goal of urban planning and existing development, striving to promote the coordinated development of society, economy, and environment through rational planning and resource utilization. However, China focuses more on solving practical problems and historical legacy issues in the urbanization process, such as the renovation of old urban areas and rural revitalization, while other countries focus more on protecting urban historical and cultural heritage and formulating long-term urban development plans. Moreover, China relies more on practical cases and field research in urban planning and existing development, emphasizing solutions to specific problems (Dang and Zhuang, 2019), while other countries tend to focus more on theoretical research and method exploration, emphasizing the formulation of long-term plans and policy measures (Rogers and Hunt, 2019).

4.5. Social infrastructure and public services

The improvement of social infrastructure and public service affects residents' quality of life and well-being. The construction of social infrastructure and the development of public services such as education, healthcare, and cultural institutions not only reflects regional and urban soft power but also serves as an important guarantee for residents' rights and interests.

In China, research on social infrastructure and public services primarily focuses on the design of elderly care service facilities, the renovation of old neighborhoods, and the aging-friendly renewal of traditional communities. Ma et al. (2020) proposed a study on the design of urban community-integrated elderly care service facilities based on social design theory, aiming to enhance the programming and design level of urban community-integrated elderly care service facilities. They emphasized that

architects should consider the relationship between people and the environment from the perspective of social design. Li and Qu (2023), based on grounded theory, studied the demand for residents' renovation of old neighborhoods involving market entities, providing key information for subsequent architectural programming and renewal designs to safeguard residents' interests. Tu et al. (2021) conducted on-site research and interviews, proposing a study on the aging-friendly renewal programming of traditional communities in large cities. They constructed a resident demand model for the renewal of old neighborhoods using group decision-making methods, providing practical references for urban public service construction.

In other countries, research on social infrastructure and public services focuses more on large-scale housing community re-programming, human-centered public space design, and mixed-use residential development. Milovanovic et al., 2023 conducted a study on the design classification of large-scale housing communities in Europe, discussing their potential for urban transformation and proposing corresponding renewal strategies. Stevens et al. (2016) studied the design of public shared spaces in large residential projects, emphasizing the need for designers to work within a more humane framework to meet the diverse needs of users for public services. Gu et al. (2023) analyzed the role of digital collaboration in mixed-use residential development projects in Australia, emphasizing the importance of digital collaboration in addressing complex programming for social infrastructure projects.

In conclusion, both China and other countries prioritize improving the quality of urban residents' lives and social welfare in their research. Secondly, they emphasize the concept of human-centered design, focusing on fully considering user needs and community participation to ensure the effectiveness and sustainability of infrastructures and public service facilities. However, in response to the same challenges, there are differences in strategies between China and other countries. The former emphasizes government-led programming and project implementation, focusing on overall planning and policy orientation, while the latter emphasizes market mechanisms and social innovation, advocating for diverse participation and win-win cooperation. Additionally, in terms of research methods and theoretical frameworks, Chinese research is more based on social science methods such as needs theory and social design theory, focusing on social perspectives, while other countries tend to use systematic and comprehensive research methods, combining multidisciplinary knowledge from architecture, urban study, anthropology, and other fields.

4.6. Architectural education and talent cultivation

Architectural education influences the future development direction and quality of the architectural industry, involving the understanding and responsiveness of the new generation of architects to societal demands. Furthermore, cultivating high-level architectural talents plays a crucial role in enhancing architectural programming and design levels and promoting innovative development in the industry.

In China, research on architectural education and talent cultivation in the field of architectural programming

primarily focuses on teaching practices in architectural design courses, the practice-oriented education model for architects, and the teaching system of architectural education. [Liang \(2019\)](#) proposed an exploration of teaching practices in architectural design courses, emphasizing the mutual confirmation and feedback mechanism of “pre-programming and post-evaluation” in design teaching, and innovatively integrated and constructed the architectural curriculum system from the perspectives of process enhancement, method construction, and spatial analysis. [Fu \(2010\)](#) discussed the training objectives of architectural programming education from the perspective of the practice model for architects, aiming to cultivate architectural professionals who meet market demands in practice. [Gao and Li \(2017\)](#) applied dynamic systems theory to analyze the dynamic development of architectural education, providing ideas and methods for positioning architectural programming disciplines in the educational system.

In other countries, research on architectural education and talent cultivation in the field of architectural programming focuses more on the development and transformation of architectural disciplines, health and well-being in architectural education, and architectural competition education to promote programming skills. [Yasar and Gür \(2022\)](#) studied the programming ideas and design methods of famous architects over the past three decades, discussing the trends in the development of architectural disciplines and changes in design paradigms, indicating that the architectural discipline retains ancient design paradigms but has changed in the handling of parameters and questioning methods. [Oliveira et al. \(2020\)](#) focused on how architecture and environmental engineering students address health and well-being issues in design projects, revealing the impact of health and well-being analysis on curriculum design in architectural education through architectural programming. [Bibbings et al. \(2018\)](#) enhanced the creativity and independent learning abilities of architectural technology students through modular teaching in actual design competition projects, promoting students’ interdisciplinary collaboration in architectural programming.

In summary, both China and other countries regard architectural education and talent cultivation as the cornerstone of industry development, aiming to enhance the quality and level of architectural professionals through innovation and improvement of the educational system. However, China focuses more on the practical skills and market adaptability of architects, emphasizing the close integration of architectural education with architectural practice to cultivate students’ abilities to solve practical problems ([Zhang and Zhuang, 2008](#)). In contrast, other countries place more emphasis on interdisciplinary collaboration and innovation in architectural education, tending to break through disciplinary boundaries, introduce new fields such as health and well-being, and cultivate students’ creativity and independent learning abilities through practical design competitions.

5. Discussion and conclusions

This study employed a mixed research method involving bibliometrics, text mining, and systematic review to review

the research status and research progress in architectural programming in China and other countries. By collecting and organizing literature related to architectural programming research from WoS and CNKI databases, this study conducts bibliometric analysis between China and other countries, including annual publication quantity, highly cited literature, authors, journals, institutions/organizations, countries/regions, co-occurrence and co-citation of keywords. We found that (1) architectural programming has become a research hotspot in academia in the past decade, but the growth in the number of related studies in China is relatively unstable compared to other countries; (2) architectural programming in China is predominantly dominated by architecture, while in other countries, it is mainly distributed across multiple disciplines such as project management, engineering construction, architecture, and urban studies; (3) research on architectural programming in China tends to adopt a diffusion perspective from inside to outside, focusing on programming theories, methods, and programming practices for various types of buildings, while other countries tend to adopt a gathering perspective from outside to inside, focusing on social impacts, building performance simulation, environmental behavior analysis, and so on.

Furthermore, this study conducted LDA topic modeling on the distribution of research topics in China and other countries to identify the research hotspots and trends in this field. We found that: The research hotspots in China mainly include programming theory and methods, intelligent design and technology, infrastructure construction, urban planning and rural revitalization, sustainable development and environmental protection, architectural education and talent cultivation, project management and decision analysis, as well as social participation and multi-stakeholder cooperation. The research hotspots in other countries mainly include project management and construction practice, digital technology and collaborative work, sustainable architectural design and practice, architectural education and professional development, urban renewal and planning decisions, as well as public service facility construction and management. Through the systematic review to compare the research progress in China with other countries, we draw the following conclusions: (1) China focuses more on the technical application and practical effects of architectural programming, while other countries focus more on interdisciplinary collaboration and innovative methods. (2) China emphasizes the practical abilities of architects and their full participation throughout the process, while other countries are more concerned with exploring critical factors and decision support in architectural programming. (3) China emphasizes government-led programming demands and project implementation, while other countries focus more on market mechanisms and social innovation. (4) China relies more on current situation research and empirical analysis, while other countries tend to adopt systematic and comprehensive theoretical research methods.

Lastly, research on architectural programming, both in China and other countries, aims not merely to refine “best practices” but to introduce transformative practices. These studies reflect two primary motivations: on one hand, some research endeavors seek to optimize existing architectural

programming processes through technological advancements and innovative practices; on the other hand, some propose entirely new theoretical frameworks and methods intended to fundamentally alter traditional architectural programming models. For example, numerous studies focus on enhancing the efficiency and quality of architectural programming through intelligent and digital means, while others attempt to redefine the core concepts and implementation paths of architectural programming through interdisciplinary approaches. In this way, we can observe the continuous evolution of the field, advancing from optimizing current practices to fostering transformative innovations.

This study has made three main contributions: (1) comprehensively reviewed and analyzed the current status and progress of architectural programming research; (2) conducted in-depth comparative analysis of research in the field of architectural programming between China and other countries; (3) introduced the latest research achievements in architectural programming and possible future research directions. Overall, this study provides a comprehensive overview of the current status of architectural programming for practitioners and scholars in the construction industry, focusing on the latest developments in this field.

However, this study has some limitations. Firstly, the scope of the literature data collection was confined to WoS and CNKI, which may not cover all relevant fields and the latest research findings. Secondly, the study limited other countries' work to English publications in the comparative analysis, which might result in an inadequate representation of the diversity and complexity of architectural programming research. Thirdly, although this study focused on the topics and trends of the literature, it did not systematically summarize the methodological flaws, data relationships, and evaluative conclusions of individual studies. Future research could incorporate literature in more languages to further explore the differences and commonalities in architectural programming across various countries and regions. Moreover, conducting more in-depth systematic reviews could more comprehensively analyze the existing gaps and potential relationships within architectural programming research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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